

SpaceTformer: a Spacetime Transformer for Photon Energy Reconstruction in the LHCb PicoCal under Pileup

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Abstract

We study photon energy regression in the proposed LHCb Upgrade II PicoCal electromagnetic calorimeter on a simulated minimum-bias sample, where the photon carries on average only 30% of its cluster’s energy. SpaceTformer, a token transformer over calorimeter cells — each token a point in detector space and time — with a physics-constrained readout, reaches an aggregate relative resolution $\sigma_{\text{eff}} = 0.0388 \pm 0.0002$, measured fully out-of-sample under a complete ten-fold cross-validation protocol (every event predicted by models that never saw it), a $3.2\times$ improvement over a six-feature gradient-boosted reference on the same sample. In a paired comparison with the encoder as the only varied component, it outperforms re-implementations of ParticleNet and GravNet in every measured cell at every seed, at $1.8\text{--}3.6\times$ their training speed — a margin that survives a per-baseline hyperparameter sweep. The main contribution is a diagnosis: the resolution ceiling was set by the input window, not the model. The starting window held under half of the innermost region’s cluster energy, and fixing the window’s size and centring cut the worst region–energy bin by 53% ($0.165 \rightarrow 0.078$), while ten encoder families, four gate-supervision protocols, five timing constructions, a capacity sweep and a recipe sweep all measured null against the same bins. A directly measured data-scaling curve, $\sigma_{\text{eff}} \propto N^{-0.18}$ over four within-protocol subsample points, predicts that tripling the simulated sample would bring the aggregate to ≈ 0.033 , leaving training data as the only open lever. The tripled-sample value is an extrapolation beyond the measured range and is labelled as such.

1 Introduction

The PicoCal is the proposed electromagnetic calorimeter for LHCb Upgrade II [18], designed to operate at instantaneous luminosities where each photon cluster is contaminated by energy from tens of simultaneous proton–proton interactions [state the simulated ν / luminosity — ask the collaboration]. In the sample used here the photon carries on average 30% of the energy inside its own cluster, with the contamination worst in the small-cell inner regions. Reconstructing the photon energy then stops being a calibration problem and becomes a separation problem.

Machine-learned regression is the natural tool, and the field has produced a family of increasingly sophisticated architectures for calorimeter and jet tasks — graph networks such as ParticleNet [1] and GravNet [2], attention models [4], and hybrids. Papers in this family usually compare against numbers published on other datasets, which makes it hard to know how much of a reported gain comes from the architecture and how much from everything around it. We instead fix one training protocol — loss, splits, seeds, readout, calibration — and vary exactly one component at a time, treating every proposed improvement as a hypothesis to be measured. Roughly 110 configurations were measured this way (Appendix A).

The thesis this discipline produced can be stated in one sentence: *on pileup-dominated PicoCal data, the resolution ceiling was set by the input window, not the model — fixing the window’s containment and centring cut the worst bin by 53%, while every architectural alternative measured null, leaving training-set size as the only open lever.* The paper presents the

headline result and its uncertainty protocol (Section 5.1), the diagnosis with its mechanism measurements (Sections 5.2–5.3), the controlled architecture campaign (Section 5.4), the ledger of measured negative results (Sections 5.5–5.7), deployment cost (Section 5.8), and the scaling case (Section 5.9).

2 Related work

Graph networks for irregular detectors. Point-cloud learning entered vision through PointNet [19] and EdgeConv [20], and particle physics through ParticleNet [1] (EdgeConv over dynamically recomputed k -nearest-neighbour graphs) and GravNet [2] (aggregation weighted by distance in a learned latent space, designed for calorimeters); the same lineage now powers full particle-flow reconstruction (MLPF [21]) and symmetry-equivariant taggers (LorentzNet [22]). ParticleNet and GravNet, the two most often recommended for our task, are re-implemented and measured here under a paired protocol (Section 5.4). We read the outcome not as a criticism of the architectures but as evidence that at $O(10^2)$ tokens with severe per-cell noise, local graph connectivity buys nothing over full attention.

Attention in particle physics. The transformer [3] entered jet physics through the Particle Transformer [4], whose pairwise physics-motivated attention bias is the direct ancestor of one of our losing arms. TimeSformer [5] showed that factorising attention along the time axis wins in video; our time-sliced attention arm transplants that idea using per-cell time as the axis, and it loses — video frames are dense and redundant along time, calorimeter timestamps are one noisy scalar per token. Patch-hierarchical attention for jets [6] motivated our ring-summary window extension (fine detail in the core, coarse summaries outside), the one transplanted idea that survives measurement. ClusTEX [7] separates global from local detector coordinates; the same separation measured here is neutral.

Set, flow, and mixing methods. Deep Sets [8] and Energy Flow Networks [9] pool per-particle embeddings additively; our EFN arm confirms that additive pooling is too weak when the task is subtractive — separating overlapping energy fields rather than summarising one. MLP-Mixer-style token mixing [10] replaces attention with fixed linear mixing across tokens; a masked Mixer arm measured here loses heavily (Section 5.4) — under pileup the informative cells move event by event, which content-based attention tracks and fixed mixing weights cannot.

Linear attention and state-space models. The Mamba lineage [11] offers linear-complexity sequence mixing whose benefit appears at long sequence length (published comparisons concern thousands of tokens). Our events carry 81–289 tokens, where full attention is exact and cheap, so we deprioritised the family without measurement and say so explicitly.

Self-supervised vision backbones. We assessed DINOv3-style pretraining [12] on cell images and declined to run it: the pretraining distribution shares no statistics with sparse calorimeter depositions, and our labelled sample (7×10^4 events) is far below the regime where frozen-backbone transfer beats direct training.

Boosted trees and calorimetry under pileup. Gradient-boosted decision trees [13, 14] remain the standard fast baseline — ATLAS calibrates electron and photon energies with tree-based regression [27] — and ours is specified in Section 3. Learned pileup mitigation began with PUMML [23] and attention-based variants [24]; learned calorimeter energy regression at CMS uses dynamic graph reduction [25] and deep clustering in the ECAL [26]. Those works subtract pileup or regress energy; ours does both in one model and reports where the ceiling comes from. The CMS HGCAL [15] programme reports the same qualitative pattern we find —

pileup-aware reconstruction gains concentrate at low energy and in the fine-granularity region — which supports reading our per-bin results as physics rather than tuning. The quantile (pinball) loss goes back to [16], its neural form to [33]; exponential weight averaging to [17]; deep ensembles to [32].

3 Dataset and task

Two simulated samples are used. The *minimum-bias sample* — the physics target and the source of every headline number — contains 72,554 photon clusters under realistic pileup, spanning the five PicoCal cell-size regions with photon energies restricted to 1–100 GeV. Its region composition (from the cross-validated half: 15 mm $\sim 12\%$, 30 mm $\sim 19\%$, 40 mm $\sim 29\%$, 60 mm $\sim 34\%$, 120 mm $\sim 6\%$) makes the aggregate metric outer-region dominated, a fact that matters repeatedly below. A *clean sample* of pileup-free photon showers from the same simulation enters only as a training auxiliary (Section 4). Each cluster provides its cells with per-cell energy, position, and two timestamps from the front and back longitudinal compartments — only $\sim 30\%$ of cells carry a usable timestamp — plus cluster-level quantities from the standard reconstruction (cluster position, summed energy, cell multiplicity). [Simulation provenance to be completed with the collaboration: generator and simulation framework versions, pileup conditions, out-of-time spillover, timestamp digitisation and resolution, photon selection.] The target is the true photon energy.

Figure of merit. The effective resolution σ_{eff} is the half-width of the minimal interval containing 68.3% of the per-event relative residuals $(E_{\text{pred}} - E_{\text{true}})/E_{\text{true}}$. It is robust to tails (Figure 7 shows why a Gaussian fit would not represent these distributions), and it is bias-blind, which is why we report linearity separately (Figure 6).

We report σ_{eff} in fifteen bins — five regions $\times$ three energy terciles, with tercile edges anchored to the starting configuration’s energy distribution per region: 46.8/73.8 GeV (15 mm), 31.7/57.5 GeV (30 mm), 19.2/34.0 GeV (40 mm), 12.1/21.5 GeV (60 mm), 9.4/16.5 GeV (120 mm) — plus the event-weighted aggregate, and continuously versus energy in Figure 4.

Evaluation protocol. The primary protocol is ten-fold cross-validation with a 5% validation slice, so each model trains on 85.5% of events. All ten folds are trained: every event in the sample is predicted exactly once, by an ensemble of five models that never saw it, with the per-event prediction the median over members. The five members per fold are three seeds of the main configuration plus one member with region-restricted recentring and one trained on all cluster cells (Section 4). During development we used a fixed 70/15/15 split with larger seed ensembles; that protocol flatters the weak bins by roughly 0.009 (15 mm low-E: 0.0691 fixed-split vs 0.0783 cross-validated), so we do not quote fixed-split numbers as headlines and we label the protocol wherever one appears. Against selection effects across the ~ 120 -configuration campaign (Appendix A): the headline protocol was fixed before the final measurement, screening runs never enter headline claims, and the half-sample preview and its correction (0.0379 $\rightarrow$ 0.0388) are reported rather than absorbed.

Uncertainties. Two kinds are quoted. On ensemble numbers: a bootstrap error over test events (400 resamples). On claims comparing configurations: the spread of σ_{eff} across independently trained members, which is the honest reproducibility scale — a pooled statistical error would assume independent test events that repeat across members. Across the 50 member-fold arms of the primary result, a single member’s aggregate σ_{eff} is 0.0397 with spread 0.0010; in the weak bins the single-member spread is 0.0070 (15 mm low-E) and 0.0046 (30 mm low-E).

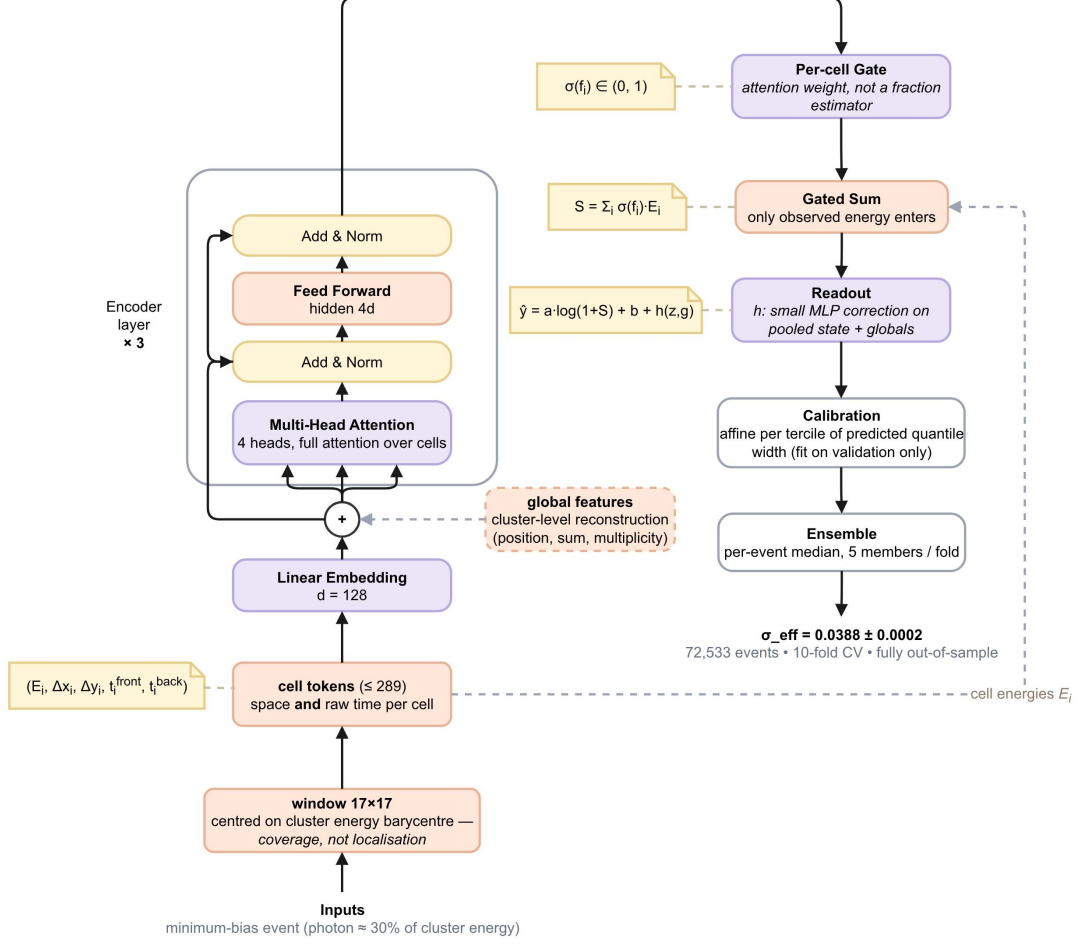


Figure 1: The SpaceFormer pipeline. Each cell is a token in detector space and time; the encoder is a plain transformer; the readout is a gated energy sum in log space; the loss is a three-quantile pinball loss with EMA weights; predictions are the per-event median over ensemble members. The two decisive components are on the far left: window size and window centring (Section 5.2).

GBDT reference. The external baseline is a histogram-based gradient-boosted regressor (300 iterations) on six summary features per cluster — log summed window energy, front/back energy ratio, cell multiplicity, log seed-cell energy, energy-weighted lateral width, and region — trained on the same events with a log-energy target, single model, fixed split. It reaches aggregate $\sigma_{\text{eff}} = 0.1253$ on this sample (per-region curves in Figure 4); the same-protocol transformer improves on it by $3.2\times$. It is a fast-baseline reference, not a tuned competitor, and we label it as such. One substitution relative to the original project plan is made explicit: the LHCb production clustering algorithms (the Cellular Automaton and its graph-clustering successor) run upstream of this dataset — the clusters we receive *are* their output — so their energy estimate enters our comparisons as the calibrated cluster sum ($\sigma_{\text{eff}} = 0.184$ on this sample, Figure 12) rather than as re-implementations.

4 Methods

Figure 1 shows the full SpaceFormer pipeline. Throughout, the *seed cell* is the highest-energy cell used by the standard reconstruction to anchor a cluster, and *ring r* denotes the square shell of cells at Chebyshev distance r from the window centre.

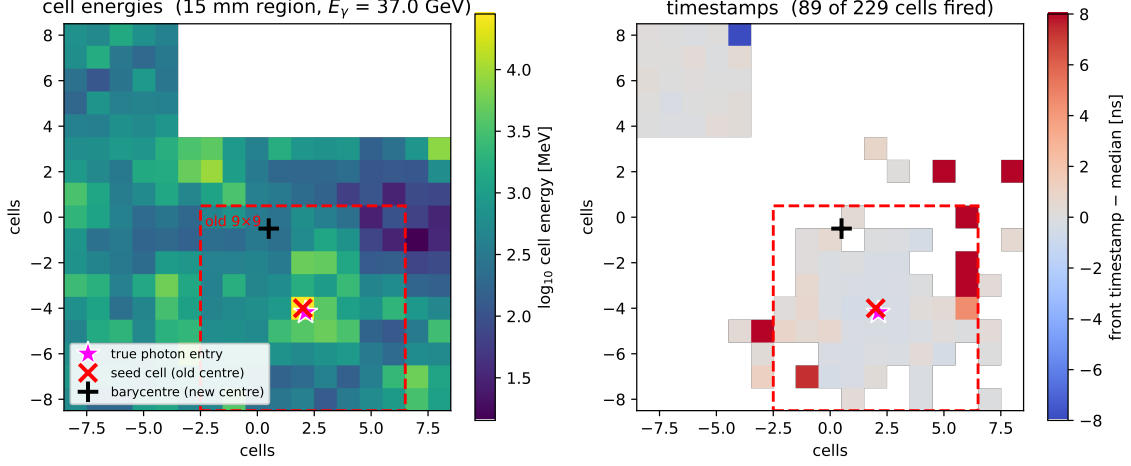


Figure 2: One real minimum-bias input to SpaceTformer: a 15 mm-region cluster ($E_\gamma = 37$ GeV, 424 cells) shown in the recentred 17×17 window. Left: per-cell energies — pileup deposits everywhere, with 74% of the cluster energy outside the starting 9×9 window (red dashed). Right: front timestamps, carried by only 39% of the fired cells. Markers: true photon entry (star), seed cell (cross, the old window centre), energy barycentre (plus, the new centre).

Tokens. Each calorimeter cell is a token carrying its energy, local coordinates relative to the window centre, and its raw front and back timestamps; cells without a usable timestamp carry explicit has-timestamp indicator channels alongside the zero-filled value, which after the linear embedding is equivalent to a learnable missing-value embedding. Cluster-level reconstruction quantities enter as global features; all inputs come from the standard reconstruction and none uses generator truth. Figure 2 shows one real input: the pileup blanket the model must subtract, the sparse timestamps, and the three centre candidates the window story of Section 5.2 turns on.

Window. The decisive design choices are *where* the window sits and *how much* it covers, not what reads it. The main configuration uses a 17×17 window centred on the cluster energy barycentre (computable from standard reconstruction quantities, hence available online) in *all* regions; one ensemble member instead restricts the recentring to the two inner regions, where the measured mis-seeding rate is highest and the gain concentrates. Section 5.2 presents the measurements behind both choices.

Encoder. SpaceTformer’s encoder is a plain token transformer [3]: $d = 128$, 4 heads, 3 layers, feed-forward width $4d$, dropout 0.1, multi-head self-attention over cells with the global features injected. The name marks what the model attends over — each token is a point in detector space *and* time, with the raw timestamps worth 20% of the aggregate resolution (Section 5.6) — not a factorised attention scheme: we measured time-factorised attention and it loses. Ten encoder families were tried against this one (Section 5.4); none won.

Physics-constrained readout. Instead of predicting the energy freely, the network emits a per-cell gate $\sigma(f_i) \in (0, 1)$ and the readout is

$$\hat{y} = a \cdot \log\left(1 + \sum_i \sigma(f_i) E_i\right) + b + h(z, g),$$

a gated energy sum in log space plus a small learned correction h (an MLP on the pooled encoder state z and the global features g). The gated sum is the dominant pathway and anchors the

prediction to observed cell energies; the correction term is free in log space and absorbs what the sum cannot express, so the constraint is an inductive bias, not a hard bound. This stabilises the low-statistics bins. We stress, and measure in Section 5.5, that the learned gate is *not* a per-cell pileup-fraction estimator and forcing it to be one hurts.

Loss and target. The choice was measured, not assumed (Table 1, all arms sharing this encoder). Predicting the energy directly and predicting a residual on the calibrated sum are indistinguishable and poor; the gated readout with a Huber loss halves the error; quantile losses beat Huber. The production loss is a pinball loss at quantiles (0.25, 0.5, 0.75) — the median is the prediction — plus a width-regularisation term that penalises wide inter-quantile intervals while softly enforcing their coverage. One arm deserves its warning label: *trimmed risk* — dropping the worst 10% or 20% of events from the loss, a textbook move for a robust target metric — is catastrophic (0.0513 and 0.0565) and unstable across seeds (± 0.05 at 15 mm low-E). The events it drops are precisely the pileup-heavy population of Section 5.3; refusing to train on them removes the hard half of the physics.

Table 1: Loss and target study, identical encoder, fixed-split protocol, aggregate σ_{eff} (five seeds unless noted).

target / loss	note	aggregate
direct energy, L2		0.0650
residual on calibrated sum, L2		0.0675
gated readout, Huber	first structural win	0.0462
gated readout, 3-quantile		0.0427
+ EMA + clean-sample auxiliary	frozen recipe	0.0390
trimmed risk, drop worst 10%	2 seeds	0.0513
trimmed risk, drop worst 20%	2 seeds	0.0565

Clean-sample auxiliary. Training batches mix the minimum-bias events with events from the pileup-free clean sample (Section 3), sharing the same loss and readout. The clean events anchor the photon response — what a shower looks like when nothing contaminates it — while the minimum-bias events teach the subtraction; removing the auxiliary costs resolution and it is part of the frozen recipe (the **CleanAux** tag in every ledger name).

Training and calibration. AdamW (learning rate 3×10^{-4} , weight decay 10^{-4}), batch 96, cosine-annealed learning rate, up to 100 epochs with early stopping on the validation loss (patience 15), and an exponential moving average of the weights (decay 0.999) [17] used for evaluation. A six-arm sweep of this recipe at the end of the project (learning rate up and down, batch up and down, coordinate jitter at two amplitudes) returned 0.0422–0.0462 against the frozen recipe’s 0.0402 on the same split — the recipe was already at its optimum. A post-hoc code audit found two schedule defects, disclosed for reproducibility: later runs inadvertently stepped the cosine schedule twice per epoch (halving its effective period), and a seventh sweep arm intended to toggle the schedule was inert and is excluded above; every comparison in this paper is between runs sharing the same code era, so no paired conclusion is affected. Post-hoc calibration is a per-group affine correction with events grouped into terciles of the predicted inter-quantile width (a per-event uncertainty proxy); under cross-validation it is fitted per fold on that fold’s own validation slice, never on pooled or test events.

Ensembling. Per-event median over five members per fold: three seeds of the main recentred configuration, one member recentring only the two inner regions, and one member trained on

all cluster cells rather than the fixed window. The median interacts correctly with the robust σ_{eff} ; a learned stacking combination was measured and is worse (Section 5.7).

5 Results

5.1 Primary result: out-of-sample resolution

Table 2 is the headline table; Figure 4 shows the same information continuously in energy against the GBDT reference, and Figure 6 shows the response linearity that σ_{eff} cannot see. All numbers are cross-validated and out-of-sample over all ten folds, five members per fold.

Table 2: Primary result: σ_{eff} per region and energy tercile, ten-fold cross-validation, all folds complete, out-of-sample (72,533 events). Parentheses give the bootstrap error on the last digits; n is the number of test events per bin (low/mid/high). “Start” is this project’s starting configuration (fixed 9×9 seed-centred window, five-seed fixed-split ensemble), shown for the two regions where it failed the per-bin target; its aggregate was 0.0390. The sample holds 72,554 clusters; the per-fold ensemble merge keys events by (energy, region) and 21 events (0.03%) collapse under identical keys.

region	low-E	mid-E	high-E	n (l/m/h)	start, low-E
15 mm	0.0783(22)	0.0451(11)	0.0340(8)	2815/2866/2685	0.1654
30 mm	0.0778(19)	0.0422(7)	0.0293(5)	4458/4828/4693	0.0957
40 mm	0.0578(10)	0.0316(5)	0.0240(4)	6977/6908/6982	
60 mm	0.0529(7)	0.0310(3)	0.0232(3)	8200/8478/8259	
120 mm	0.0642(22)	0.0367(11)	0.0291(8)	1546/1528/1310	
aggregate	0.0388 ± 0.0002			72533	0.0390

Every bin is below 0.079; thirteen of fifteen are below 0.065. The two weakest bins are the low-energy bins of the two innermost regions, where pileup contamination is worst and event counts smallest. Figure 3 shows all fifteen bins against the starting configuration: the gains concentrate exactly where the containment measurement (Section 5.2) says they should — the inner-region low-energy bins ($0.165 \rightarrow 0.078$ and $0.096 \rightarrow 0.078$) — with small regressions in the outer bins that were already contained, the largest at 120 mm low-E ($0.0529 \rightarrow 0.0642$; the starting value coincides numerically with the 60 mm low-E entry above it). At 120 mm the barycentre is dragged by pileup tails while the seed cell was already correct, so recentring is measured to cost resolution there — yet four of the five ensemble members use it; the fifth, region-restricted member limits the damage.

Figure 5 fits the standard calorimeter parameterisation $\sigma_{\text{eff}}(E) = a/\sqrt{E} \oplus b/E \oplus c$ to the cross-validated points per region. The fit tells the physics story compactly: the $1/E$ term — which for a pileup-dominated sample plays the role of the noise term, an energy-independent contamination in GeV — is largest by far in the 15 mm region ($b = 2.1$ GeV against 0.16–0.34 GeV in the outer regions), while the constant terms sit below 2.5% everywhere. The residual low-energy weakness of the inner regions is pileup energy, not sampling statistics.

5.2 The diagnosis: window containment and window centring

The improvement over the starting configuration came from two reconstruction-level corrections, and this subsection presents the measurements that found them, in the order they happened.

The window half-width had been fixed at $w=4$ (9×9 cells) from the start of the project and never varied. Figure 8 shows the measurement that broke this: the mean contained fraction of cluster energy versus window half-width, per region. At $w=4$ the seed-centred window holds 41% of the 15 mm cluster’s energy against $\geq 92\%$ in the three outer regions — one shared

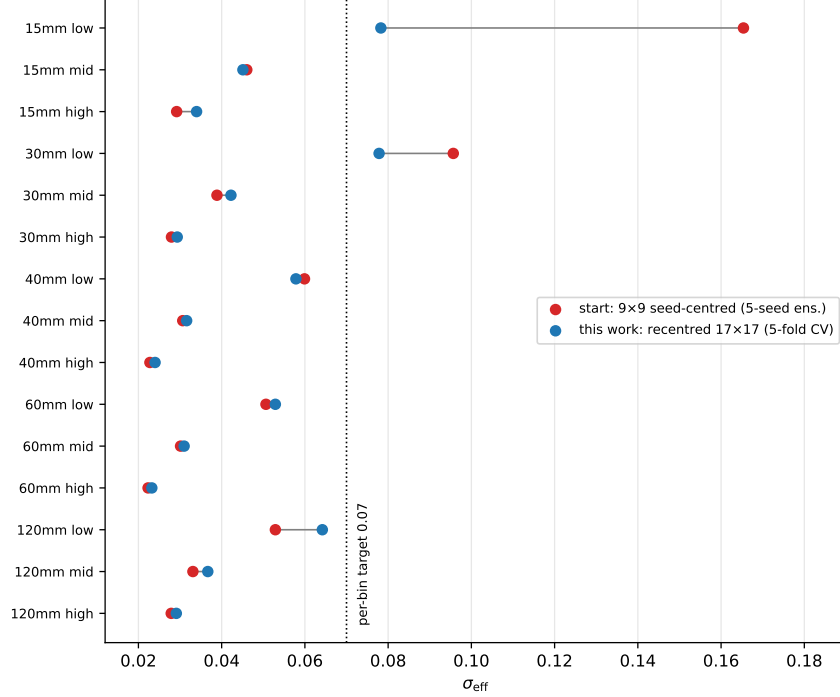


Figure 3: All fifteen region–energy bins: starting configuration (red, fixed-split five-seed ensemble) versus this work (blue, cross-validated ensemble). The two protocols differ slightly (Section 3); the dominant moves — 15 mm low $0.165 \rightarrow 0.078$, 30 mm low $0.096 \rightarrow 0.078$ — dwarf that difference.

window width had been silently tuned for the abundant outer regions. The physics behind the asymmetry: the shower has a single physical width (99.7–100% of photon energy within 120 mm in every region on the clean sample), so its width *in cells* varies eightfold across regions.

Sweeping the seed-centred window from 5×5 to 21×21 (Figure 9) confirms the diagnosis in resolution space: 15 mm low-E falls monotonically from 0.186 to 0.102 as the window opens — the single largest lever found in the whole project — while the *aggregate barely moves* (0.038–0.044 across the whole scan), because the outer regions dominate the event count and were already contained. That flatness is exactly how the defect stayed invisible: the headline metric hid it. Ring summaries (coarse sums of the cells outside the window, after [6]) let $w=8$ match $w=10$ at half the tokens, but only extend a resolved core; at $w=4$ they recover little of what truncation destroys.

The second correction is *where* the window sits. Re-centring it on the cluster energy barycentre gives the biggest single step in the cumulative decomposition (Table 3: $0.1156 \rightarrow 0.0711$), and it invites a wrong story — that the barycentre finds the photon better than the seed cell. That story is measurably false. Figure 10 shows the distance of each candidate centre from the true photon entry point in the 15 mm region: the seed cell is *much closer* to the photon (median 0.44 cells against 3.06), yet the barycentre-centred window wins. The window is not a photon-pointer; it is an aperture, and what matters (Figure 8, solid vs dashed) is that the aperture covers the energy field — photon tail plus the pileup the model must subtract — and gives the network a coordinate frame anchored to that field rather than to one volatile cell. (The decomposition in Table 3 ends at the cross-validated 0.0783.) Centre estimators aimed at better photon localisation (an argmax-iterated log-centroid, a hybrid that follows the seed except on disagreement, a clustered 3×3 seed) were all measured worse than the plain barycentre.

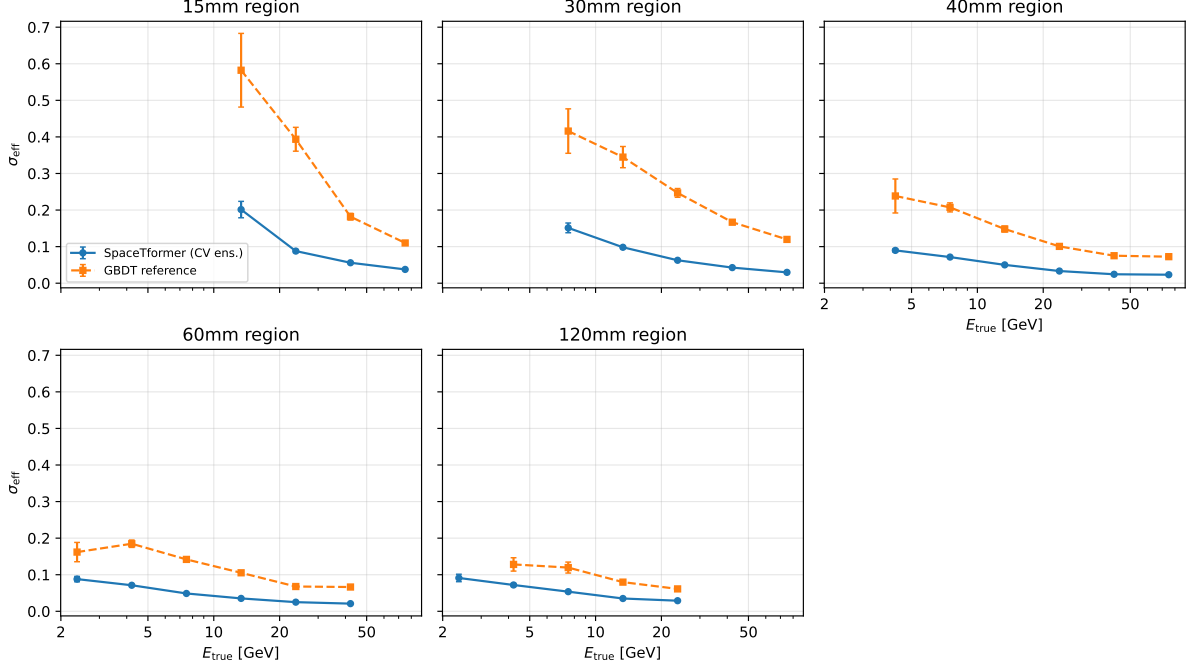


Figure 4: σ_{eff} versus true photon energy per region: SpaceTformer (cross-validated ensemble, circles) against the six-feature GBDT reference (single model, fixed split, squares). Error bars are bootstrap errors over test events. The gain is largest at low energy and in the inner regions, matching the pileup-contamination pattern.

Table 3: Cumulative decomposition of σ_{eff} in the 15 mm low-energy bin. The first three rows use the development protocol (fixed split, five-seed ensemble); the last row is the out-of-sample cross-validated number, higher because its per-fold ensembles are smaller — the two protocols are not directly comparable and both are shown deliberately.

step	15 mm low-E	protocol	mechanism
9×9 seed-centred window	0.1654	fixed split	41% containment (Fig. 8)
widen to 17×17	0.1156	fixed split	coverage
+ re-centre on barycentre	0.0711	fixed split	coverage again (see text)
cross-validated ensemble	0.0783(22)	CV	headline protocol

5.3 Anatomy of the remaining error

With the window fixed, we dissected what is left, joining per-event truth diagnostics to the best ensemble’s residuals (5,372 joined events; diagnostics are truth-based and used only for analysis, never as inputs). Four measurements structure the remainder.

(1) **The core width is pileup magnitude, not a missing mechanism.** Splitting each weak bin at the median of the ratio of window energy to true energy — a direct measure of how much non-photon energy the window holds — separates two populations (Table 4). Every low-energy bin tested, including 120 mm which was never window-limited or mis-seeded, is a mixture of a clean population already at 0.04–0.07 and a pileup-heavy population near 0.10. The inputs that describe pileup magnitude (window-to-cluster energy ratio, occupancy, density) are already features, so the model knows which population an event belongs to and still cannot delete the fluctuation: this is the aleatoric component, localised. It is what motivates reading the residual as a statistics problem (Section 5.9) rather than a modelling one.

(2) **The tails are catastrophic and separate from the core.** The ratio of the 95% to

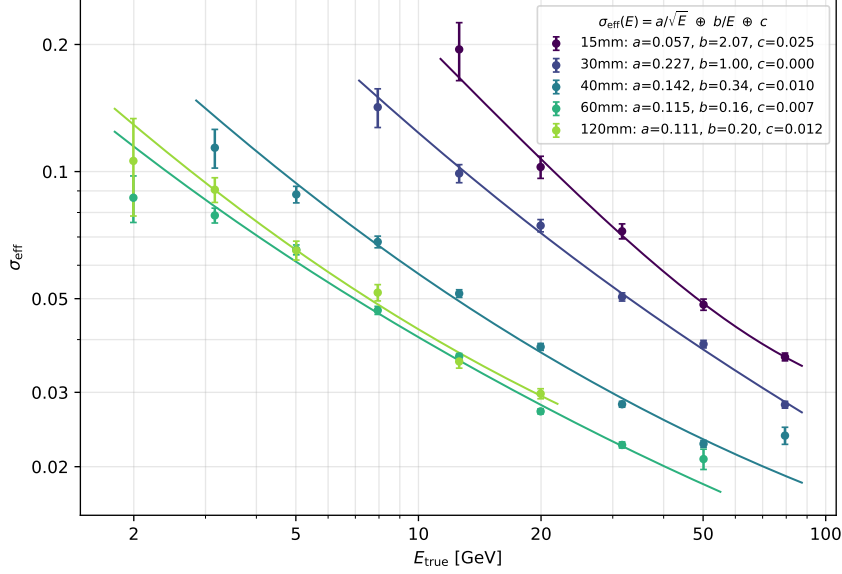


Figure 5: σ_{eff} versus true energy per region (cross-validated ensemble, bootstrap errors) with the standard parameterisation $\sigma_{\text{eff}}(E) = a/\sqrt{E} \oplus b/E \oplus c$ fitted per region. The b (noise) term absorbs the energy-independent pileup contamination and is an order of magnitude larger in the 15 mm region than in the outer ones.

Table 4: σ_{eff} of each low-energy bin split at the median per-event pileup contamination (truth-based diagnostic). The pileup-heavy half sits near 0.10 in every region — including 120 mm, which never had a window or centring problem.

bin	cleaner half	pileup-heavy half
15 mm low-E	0.0683	0.0895
30 mm low-E	0.0598	0.0977
120 mm low-E	0.0370	0.0977

the 68% residual quantile is ≈ 5 in all three bins of Table 4, against 2.0 for a Gaussian: a 5–10% population with $|\text{res}| \sim 0.4\text{--}0.6$ sits far outside the core (Figure 7). σ_{eff} barely sees these events, so no tail-directed intervention moves the headline number — but they matter for physics use, and their likely origin (wrong-photon matches, windows dominated by an overlapping cluster) is a reconstruction question upstream of regression. For deployment, the model already carries a per-event confidence: the predicted inter-quantile width used by the calibration (Section 4) widens on exactly these events and can flag them at inference without any additional head.

(3) Ambiguity is measured, priced in, and still costs. After recentring, 30 mm low-E events where the barycentre and the seed cell agree within one cell resolve at 0.0489; the rest at 0.0796. The disagreement magnitude is observable at inference and is already an input — the model knows which events are ambiguous and the gap persists, which again says magnitude of ambiguity, not missing information.

(4) Timing availability correlates the right way. Events with more time-stamped cells do better (15 mm low-E: 0.0706 with above-median timing coverage vs 0.0823 below), consistent with the timing ablation (Section 5.6) — denser timing coverage in the detector design would feed this model directly.

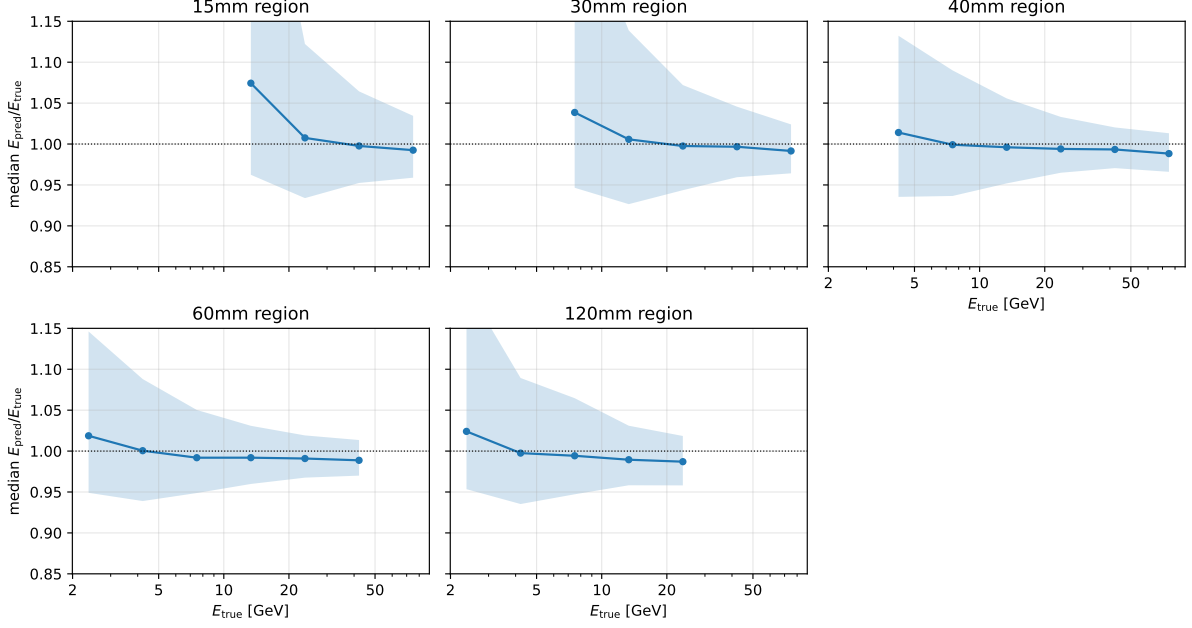


Figure 6: Response linearity: median $E_{\text{pred}}/E_{\text{true}}$ versus true energy per region for the cross-validated ensemble, with the 16–84% band shaded. σ_{eff} is bias-blind, so this plot carries the scale information the resolution table cannot.

5.4 Architectures: ten families, one survivor

The architecture campaign consumed the most compute and produced the cleanest null. Ten alternatives to the plain transformer were measured behind the same readout, loss and splits; several are re-implementations of published architectures and several were our own designs, built on physics arguments we found convincing at the time. None won.

The sharpest comparison is the paired one (Table 5): ParticleNet (EdgeConv with dynamic k NN) and GravNet re-implemented behind the identical readout, loss, splits and window, so the encoder is the only difference, with two seeds per baseline and five for the transformer. SpaceTformer wins every cell at every seed. One fairness caveat is owed: the baselines inherit our frozen training recipe, which was validated on the transformer; they received their published architectural defaults but no dedicated hyperparameter sweep, and the comparison should be read with that asymmetry in mind. A dedicated six-arm sweep (learning rate halved and doubled, and neighbourhood size k raised to 24, per baseline) closes this caveat, and we report it in full: every tuned arm still trails. ParticleNet reads 0.0456/0.0408/0.0422 (lr 10^{-4} / lr 6×10^{-4} / $k=24$) and GravNet 0.0443/0.0455/0.0427. The best of the six, ParticleNet at lr 6×10^{-4} , narrows the aggregate gap to +0.0009 (about twice the transformer’s seed spread) and reaches 0.0785 at 15 mm low-E (within the transformer’s spread there) — but remains +0.0045 behind at 30 mm low-E, five times the spread, at $3.6\times$ the training cost. The robust statement is therefore: no tuned baseline matches the transformer at equal cost, and none matches it in the 30 mm weak bin at any cost we measured. The aggregate gaps of the paired default comparison (+0.0016 ParticleNet, +0.0038 GravNet against the transformer’s five-seed mean 0.0399) exceed the transformer’s own seed spread of 0.0005; the weak-bin gaps are larger still and consistent across bins, baselines and seeds.

Table 6 is the full campaign. “Era” matters: early arms ran before the window and recentring fixes, against the matching reference of their time.

Three failures are informative enough to deserve a sentence each. The *time-sliced attention* arm was our own flagship design — bucket cells by time pull, attend within buckets, exchange pooled summaries — built on the observation that pileup energy has a 13 ns time spread at 15 mm

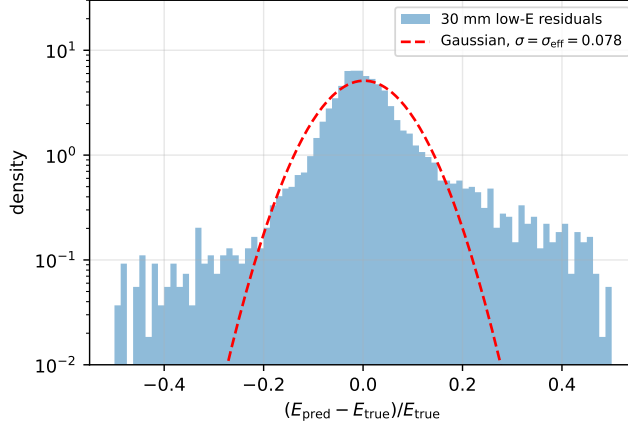


Figure 7: Relative residual distribution in the 30 mm low-energy bin (cross-validated ensemble, log scale) against a Gaussian of width σ_{eff} . The non-Gaussian tails are why σ_{eff} is the figure of merit and why tail-resaping interventions do not move it.

Table 5: Head-to-head σ_{eff} under an identical fixed-split protocol. Single models, no ensembling, which is why the transformer row differs from Table 2’s ensemble numbers; mean $\pm$ spread across seeds (five for SpaceTformer, two for each baseline). Training time is per model on one A100.

encoder	aggregate	15 mm low-E	30 mm low-E	time
SpaceTformer (ours)	0.0399 $\pm$ 0.0005	0.0759 $\pm$ 0.0058	0.0768 $\pm$ 0.0009	33 min
ParticleNet	0.0415 $\pm$ 0.0004	0.0815 $\pm$ 0.0003	0.0872 $\pm$ 0.0053	120 min
GravNet	0.0437 $\pm$ 0.0006	0.0861 $\pm$ 0.0045	0.0922 $\pm$ 0.0040	60 min

against 1.6 ns for the photon. It lost twice, on two eras of the pipeline, because slicing removes exactly the cross-time comparisons the full-attention model uses. The *grid CNN* represents the event as a dense image; it loses 20% to the token view of the same cells — under pileup, most of the grid is noise, and convolution must spend capacity learning to ignore what tokenisation simply omits. The *pairwise-bias* arm imports the Particle Transformer’s physics-motivated attention bias, which helps on jets with hundreds of particles and well-defined pairwise invariants; at 81 noisy cells it only constrains the attention away from what the data wanted. The *masked MLP-Mixer* replaces attention with fixed learned mixing weights across token positions; it loses 63% (0.0654 vs 0.0402) because under pileup the informative cells move event by event — exactly the situation content-based attention handles and position-based mixing cannot. Capacity is not the explanation for any of this. Doubling the transformer to $d=256/6$ layers collapses under the frozen recipe (0.1179); we diagnosed the collapse directly, and it is an optimization artefact, not saturation: learning-rate warmup alone recovers to 0.0692, pre-layer normalisation [31] to 0.0474, and both together restore full trainability at 0.0412 — which still does not beat $d=128$ ’s 0.0402. Halving to $d=64/2$ layers costs 15% (0.0463). The conclusion survives the diagnosis in a stronger form: the large model *trains* and still buys nothing. At 81–289 tokens per event, full attention is not the bottleneck, and nothing cheaper matches it.

5.5 The gate is not a fraction estimator

The readout’s per-cell gate looks like it should estimate the photon fraction of each cell, yet on a paired overlay sample with per-cell truth labels its correlation with the true fraction is only 0.211. We measured four ways of “fixing” this: direct supervision of the gate on overlay truth, an auxiliary fraction head, per-cell prior features from a fitted estimator, and teacher distillation from that estimator. All four are neutral or worse. Meanwhile a gradient-boosted per-cell

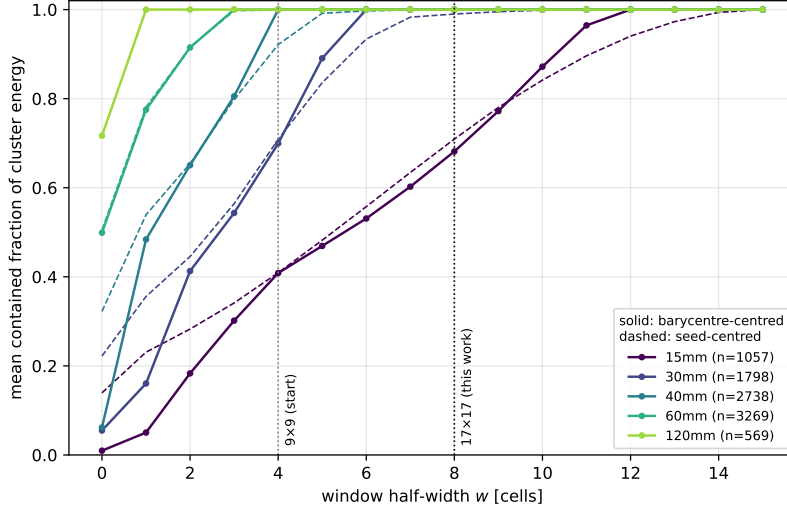


Figure 8: Mean contained fraction of cluster energy versus window half-width w , per region, on the minimum-bias sample (12 files, 9,431 events), for barycentre-centred (solid) and seed-centred (dashed) windows. The vertical lines mark the starting 9×9 window and this work’s 17×17 . At $w=4$ the 15 mm region is the only one below half containment.

estimator given the identical per-cell observables reaches correlation 0.945 on the same sample — the information is present in the inputs, and the network routes it around the gate rather than through it. The gate is a learned attention weight inside an integral, not a physical quantity, and constraining it to be physical costs resolution. We flag this because the “interpretable gate” framing is publication-ready and, on this measurement, wrong.

The gate study sat inside a larger arc on *overlay supervision*: manufacturing per-cell truth by overlaying clean photon showers onto pileup and training against those labels. Its structural half worked — an explicit subtract-then-calibrate decomposition, where one branch estimates the pileup field and a second calibrates what remains, reached 0.0465 as an ensemble in its era — but the supervision half did not: training the subtraction branch on overlay labels was neutral against letting it learn freely. A domain measurement explains why: synthetically overlaid events are distinguishable from real minimum-bias events by a classifier at AUC 0.93 on cell-level observables, so labels manufactured by overlaying do not come from the distribution the model must serve. Structure helped; manufactured labels did not.

5.6 Timing

Per-cell timing is the physics handle expected to matter under pileup, and Figure 11 shows it is a *pileup* handle specifically. Retraining the starting configuration with both timestamps removed gives aggregate 0.0485 against 0.0390, 15 mm low-E 0.2033 against 0.1654, and 30 mm low-E 0.1556 against 0.0957 — timing carries 20% of the aggregate resolution and up to 38% in the weak bins — while the identical ablation on the pileup-free clean sample is neutral (0.0476 with vs 0.0460 without). The timestamps carry no energy information; their entire value is separating overlapping sources in time

[state the simulated timestamp resolution once confirmed from the simulation configuration]. But every *engineered* use of timing loses to handing the network the raw timestamps: three time-pull coordinate constructions (per-cell time scaled by its expected spread), hard timing cuts, time-sliced attention, an inverse-variance front/back combination, and a Fourier basis on the raw time channels (0.0413 vs 0.0402) were all measured and all reduce or at best match. The network already extracts what the timestamps carry; pre-digesting them removes information.

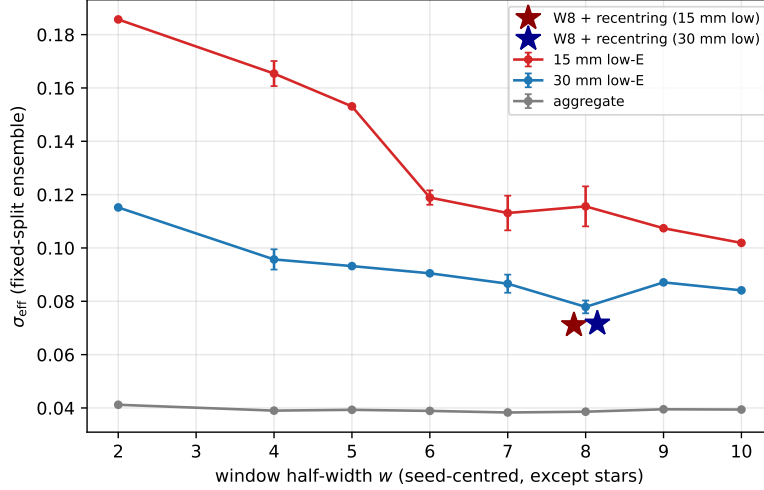


Figure 9: The window scan: σ_{eff} versus window half-width, seed-centred (circles; fixed-split ensembles, error bars where multi-seed), with the recentred $w=8$ configuration as stars. The aggregate is flat while the worst bin halves: the headline metric hid the defect.

The result independently supports the case for dedicated precision-timing layers at the HL-LHC [28]: every picosecond of per-cell timing resolution feeds a model like this one directly.

5.7 Closed doors

Table 7 is the ledger of measured nulls beyond the encoder campaign. Each row is a mechanism a reviewer — or we ourselves — would otherwise propose. For scale: the single-member spread on the aggregate is 0.0010 and the bootstrap error on the ensemble aggregate is 0.0002 (Section 3); rows marked “worse” exceed it, rows marked “neutral” sit within it.

Beyond these, fourteen per-cell and global feature mechanisms were measured on the same protocol. The ones that stayed: extra reconstruction globals, local density, and the clean-sample auxiliary. The ones measured neutral-or-worse, each with its row in Appendix A: time-pull coordinates (+0.017 at 15 mm low-E), orthogonal-residual coordinates, absolute time, local energy fraction, depth ratio, gate-side global injection, physics/occupancy summaries, FiLM conditioning on pileup context, signed gates (+0.009 at 15 mm low-E on the recentred configuration), trainable gate temperature, low-energy cell reweighting at four strengths, and millimetre-unit geometry (neutral on the final configuration; its one win pre-recentring was superseded by recentring itself). The consistent reading across all fourteen: the token set already carries the information, and the model finds it without hand-built coordinates.

5.8 Computational cost

A reconstruction algorithm is only deployable at a cost, so we benchmark inference throughput on commodity hardware (laptop CPU, i9-13900HX, batch 64, median of 5 timed passes on an idle machine). Figure 12 places each model family on the accuracy–throughput plane. At the 9×9 window a single transformer runs at $\sim 5,000$ clusters/s (0.20 ms per cluster) and the five-seed ensemble at 992 clusters/s; the GBDT reference is $3.3\times$ faster per cluster than a single transformer pass and $3.1\times$ worse in resolution; the analytic calibrated sum is effectively free (5.6×10^6 clusters/s) and $4.6\times$ worse. The final 17×17 recentred configuration pays the quadratic attention cost of 289 vs 81 tokens: 555 clusters/s per model, 111 clusters/s for the five-member ensemble, measured on the same harness. On one H100 GPU the same configuration reaches 1,017 clusters/s at batch 1 (latency-bound) and 41,345 clusters/s at batch 1024 — $24 \mu\text{s}$

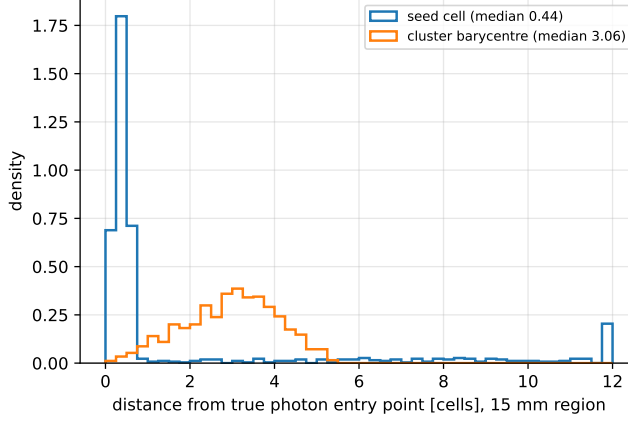


Figure 10: Distance of the two window-centre candidates from the true photon entry point, 15 mm region. The seed cell is the better photon locator by $7\times$ in the median, yet centring on the barycentre gives the better resolution — direct evidence that the window’s job is coverage of the energy field, not photon localisation.

Table 6: All encoder families measured, fixed-split protocol, aggregate σ_{eff} with the era’s matching transformer reference; single-seed values, which is why the ParticleNet/GravNet rows differ slightly from Table 5’s two-seed means. The two time-factorised arms are one family measured in two eras. Screening = cut before a five-seed run.

encoder	lineage	aggregate	reference	outcome
pair transformer	pairwise tokens	0.0627	0.039–0.041	lost, early era
energy-flow network	Deep Sets / EFN [9]	0.0623	0.039–0.041	lost, early era
spacetime-factorised (v1)	TimeSformer [5]	0.0543	0.039–0.041	lost, early era
geometric pairwise-bias	ParT [4]	0.0420	0.0390	lost
grid CNN	dense image	0.0476	0.039–0.040	lost (2 seeds)
ConvNeXt token mixer	modern CNN	—	—	cut at screening
patch tokens (ViT-style)	[6]	—	—	cut at screening
time-sliced attention (v2)	TimeSformer, on final config	0.0442	0.0402	lost
GravNet	[2]	0.0432	0.0402	lost, paired
ParticleNet	[1]	0.0412	0.0402	lost, paired
masked MLP-Mixer	[10]	0.0654	0.0402	lost
token transformer	[3]	0.0390		the survivor

per cluster for a single model, $121\mu\text{s}$ for the five-member ensemble — which brings batched inference into the regime where a conversation about high-level-trigger deployment becomes quantitative (`reports/benchmark_gpu.csv`). Accuracy per compute is therefore a real trade-off the experiment can choose along: from the free analytic sum to the full ensemble spans four orders of magnitude in throughput for a $4.7\times$ span in resolution.

5.9 Scaling: the remaining lever

With representation, architecture, supervision, recipe, ensembling and calibration each closed by measurement, the remaining open lever is training-set size, and we measured its slope directly: the final configuration retrained on deterministic 25%, 50% and 75% subsamples of the training set (validation and test untouched, same recipe, three seeds per point) plus the full-data point (Figure 13). The four points are strikingly linear in log-log and give

$$\sigma_{\text{eff}} \propto N^{-0.18},$$

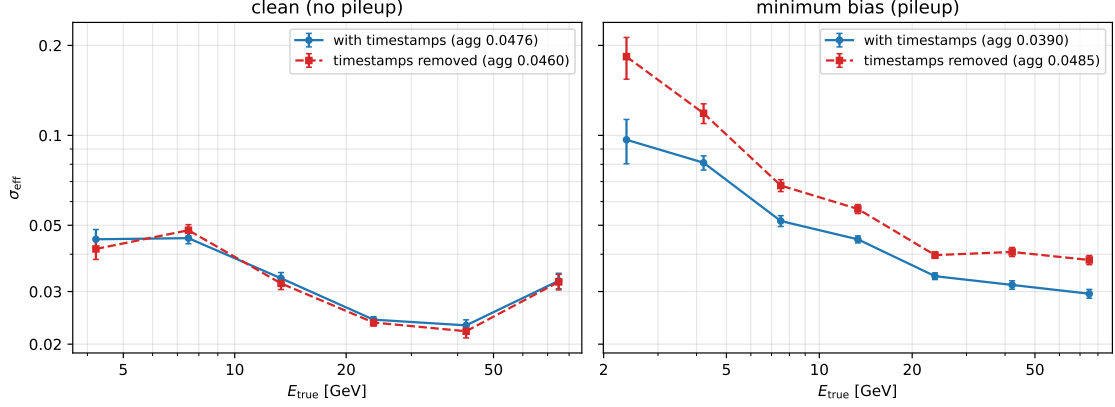


Figure 11: The timing ablation, side by side on the pileup-free clean sample (left; two-seed ensembles, earlier-era configuration) and the minimum-bias sample (right; the starting configuration). Removing the per-cell timestamps is neutral without pileup and costs 20% aggregate — concentrated at low energy — under pileup: timing is a pileup-separation tool, not an energy observable.

Table 7: Negative results, all measured under the frozen protocol. Aggregate σ_{eff} comparisons are against the matching-protocol value 0.0379 (the half-sample CV ensemble current when each was measured) or 0.0402 (fixed-split single seed) as labelled.

family	protocols tried	outcome
gate supervision	4 (direct, aux head, prior feature, distill)	neutral or worse
engineered timing	5 (pulls $\times 3$, cuts, slices, front/back comb.)	all worse
training recipe	lr $\times 2$, batch $\times 2$, jitter $\times 2$	0.0422–0.0462 vs 0.0402
ensembling	learned stacking on held-out data	0.0414 vs 0.0379, worse
recalibration	post-hoc per (region $\times$ E-decile)	0.0386 vs 0.0379, worse
test-time augmentation	D4 averaging	neutral, $8\times$ cost
capacity	$d=256/6L$; $d=64/2L$	0.1179 / 0.0463, both worse
capacity, diagnosed	+ warmup; + pre-LN [31]; + both	0.0692; 0.0474; 0.0412 — trains, no gain
time representation	Fourier basis on raw time channels	0.0413 vs 0.0402, worse

identically for single models ($0.0524 \rightarrow 0.0466 \rightarrow 0.0439 \rightarrow 0.0402$) and for three-seed ensembles (exponent -0.179), and the worst bin scales in parallel (15 mm low-E: $0.1254 \rightarrow 0.0729$ across the same points). This within-protocol curve supersedes two earlier confounded two-point estimates (-0.13 and -0.28) and lands between them; power-law scaling of error with dataset size is the generic empirical behaviour of deep networks [30, 29], so the linearity is expected even if the exponent is task-specific. Extrapolated threefold — an extrapolation, and labelled as such — it predicts an aggregate of ≈ 0.033 . The qualitative case is independent of the fit: every intervention that did not add training data saturated at aggregate ≈ 0.039 , and the residual error is dominated by per-event pileup magnitude (Section 5.3), a statistical quantity that more data smooths.

6 Limitations and future work

Four limitations bound the claims. (1) Everything is measured on one simulated sample of one detector concept; the diagnosis method transfers, the numbers may not. Two internal transfers partially mitigate this: the pileup-free clean sample is a second distribution on which the timing conclusion reproduces with the opposite sign (Figure 11), and the window diagnosis holds across five detector regions whose granularity spans a factor of eight — effectively five

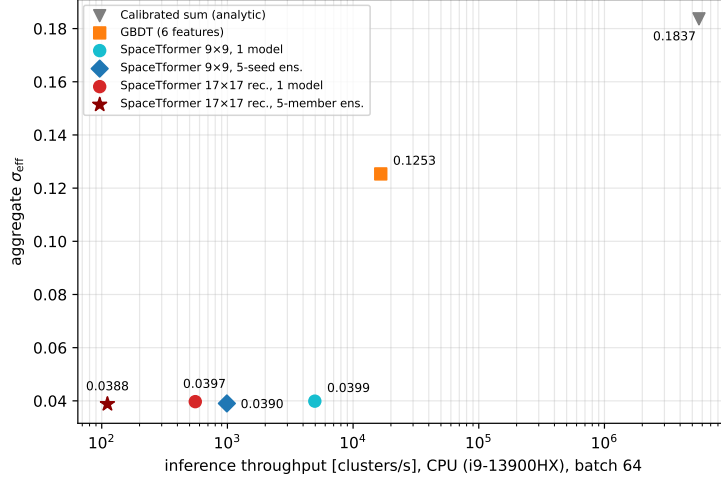


Figure 12: Aggregate σ_{eff} against inference throughput (CPU, batch 64; log scale). The four left points use the fixed-split protocol; the two 17×17 points use the single-member mean (0.0397) and the cross-validated ensemble (0.0388), with the forward-pass rate measured at 289 tokens on the same harness. The transformer buys its resolution by reading cells individually, and the window that fixed the weak bins costs a further $9 \times$ in attention compute.

related sub-detectors. (2) The architecture comparison is paired and now includes a per-baseline hyperparameter sweep (learning rate, neighbourhood size), but the sweep is small; its conclusion is about this task’s regime, not about the architectures in general. (3) The scaling prediction at $3 \times$ data is an extrapolation beyond the measured range of the (otherwise clean) four-point curve. (4) The half-sample preview of the primary table read 0.0379 aggregate; the complete ten-fold table reads 0.0388 — a real lesson in why half-done cross-validation must not be quoted as final, which we record here deliberately.

On venue fit, we state the contribution class plainly: this paper offers a measurement-driven diagnosis, a controlled architecture comparison, and a ledger of verified negative results — not a new architecture; it should be evaluated as such.

Three directions follow directly from the measurements. The catastrophic tails (Section 5.3) are a reconstruction problem — wrong-photon matches and overlapping clusters — addressable upstream of any regressor. The timing-coverage correlation suggests that denser timestamp coverage is a detector-design lever this model would convert directly into resolution. And the data request is now quantitative: tripling the minimum-bias simulation is predicted, with stated caveats, to bring the aggregate below 0.035.

7 Conclusions

On a simulated minimum-bias PicoCal sample, SpaceTformer reaches $\sigma_{\text{eff}} = 0.0388 \pm 0.0002$ aggregate, fully out-of-sample over the complete ten-fold protocol — a $3.2 \times$ improvement over a six-feature GBDT reference — and beats paired re-implementations of ParticleNet and GravNet in every measured cell at every seed, at a fraction of their cost. The improvement over our own starting point — 53% in the worst region–energy bin — came entirely from measuring and fixing the input representation: window containment and window centring. Everything else we tried is reported with its numbers in Sections 5.4–5.7 and Appendix A, so it does not have to be re-tried silently. The measured scaling curve, $\sigma_{\text{eff}} \propto N^{-0.18}$, makes the path below $\sigma_{\text{eff}} = 0.035$ a data request rather than an architecture search.

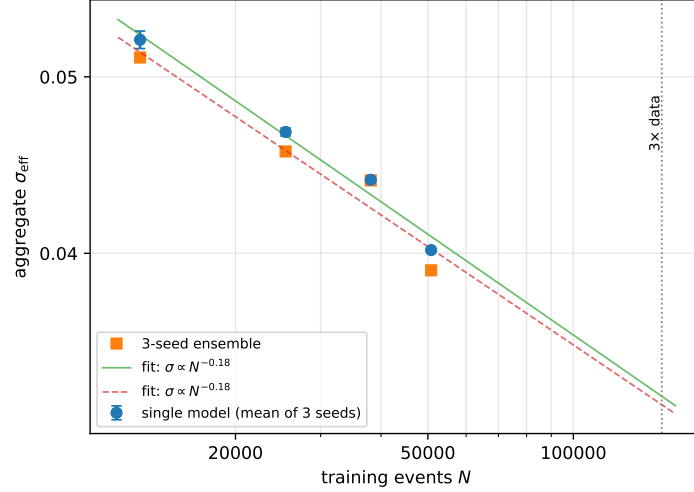


Figure 13: The measured data-scaling curve: aggregate σ_{eff} of the final configuration retrained on 25/50/75/100% of the training set (fixed-split protocol, three seeds per point; the 75% point completed one seed). Both the single-model mean and the three-seed ensemble fit $\sigma_{\text{eff}} \propto N^{-0.18}$; the dotted line marks a tripled sample, at a predicted aggregate of ≈ 0.033 .

Acknowledgements

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Code and data availability

All experiments run from a single training script with a flag for every mechanism in this paper; the prediction CSVs behind every reported number and the scripts that generate every figure are versioned alongside the code at <https://github.com/Lworakan/GSoC2026-pical-spacetime-transformer>. The simulated samples belong to the LHCb collaboration and are not redistributed.

A Complete experiment ledger

Table 8 lists every configuration with a saved prediction set on the minimum-bias sample, scored by the same seed-ensemble σ_{eff} (median prediction per event across seeds, then one σ_{eff}). The table is generated from the prediction files by the same script that produced every number in the body; nothing is transcribed by hand. Configuration names compose tags: **Wn** window half-width, **CleanAux** clean-sample auxiliary, **Ex** extra globals, **Dn** density, **Qd** quantile loss with width term, **Ema** weight averaging, **Rc** recentred window, **Rr01** recentring restricted to the 15/30 mm regions, **Ac** all cluster cells, **Kn** cross-validation fold n , **Rn** ring summaries to ring n , **Tp** time pulls, **Gs/Pf/Pt** gate supervision variants, **Mm** millimetre geometry, **Tc** front/back time combination, **Lr/B/Cos/Cj** recipe-sweep arms, **Tr** trimmed risk, **NoTime** timestamp ablation.

Table 8: Complete experiment ledger: every configuration with a saved prediction set on the minimum-bias sample, scored identically (seed-ensemble σ_{eff} ; $\pm$ is the spread across seeds). Configuration names are the run identifiers from the training script; one- and two-seed rows are screening runs.

configuration	seeds	aggregate	15 mm low-E	30 mm low-E
BDT	1	0.1253	0.3453	0.3039
CalibratedSum	1	0.1837	0.5955	0.5320
EfnResidual	5	0.0623	0.2969 ± 0.0059	0.2021 ± 0.0068
GateHuber	5	0.0462	0.2157 ± 0.0095	0.1667 ± 0.0082
MeanDirect	5	0.0650	0.2946 ± 0.0089	0.2202 ± 0.0054
MeanResidual	5	0.0675	0.2885 ± 0.0103	0.2259 ± 0.0097
PairT	5	0.0627	0.2936 ± 0.0093	0.2186 ± 0.0056
Spacetime	5	0.0543	0.2429 ± 0.0099	0.1871 ± 0.0052
SubEnsembleTTA	1	0.0440	0.1721	0.1059
SubNetW10CleanAuxExDnQdEma	1	0.0394	0.1019	0.0841
SubNetW2CleanAuxExDnQdEma	1	0.0412	0.1857	0.1152
SubNetW2CleanAuxExDnTpQdEma	1	0.0420	0.1752	0.1253
SubNetW4	5	0.0443	0.1838 ± 0.0156	0.1102 ± 0.0036
SubNetW4CleanAux	2	0.0451	0.1656 ± 0.0002	0.1108 ± 0.0002
SubNetW4CleanAuxD4QdEma	1	0.0416	0.1696	0.1071
SubNetW4CleanAuxDnQdEma	2	0.0416	0.1661 ± 0.0064	0.1025 ± 0.0009
SubNetW4CleanAuxExDnAbsQdEma	1	0.0407	0.1652	0.1030
SubNetW4CleanAuxExDnAuxQdEma	2	0.0395	0.1646 ± 0.0004	0.1016 ± 0.0030
SubNetW4CleanAuxExDnDepQdEma	2	0.0391	0.1724 ± 0.0017	0.0994 ± 0.0024
SubNetW4CleanAuxExDnF6QdEma	2	0.0401	0.1820 ± 0.0123	0.1187 ± 0.0035
SubNetW4CleanAuxExDnFmQdEma	1	0.0406	0.1621	0.0984
SubNetW4CleanAuxExDnGs10PtQdEma	2	0.0410	0.1704 ± 0.0082	0.1186 ± 0.0058
SubNetW4CleanAuxExDnGs10QdEma	2	0.0386	0.1526 ± 0.0056	0.0939 ± 0.0039
SubNetW4CleanAuxExDnGs50PtQdEma	2	0.0421	0.1858 ± 0.0145	0.1209 ± 0.0007
SubNetW4CleanAuxExDnGs50QdEma	2	0.0383	0.1686 ± 0.0043	0.0950 ± 0.0026
SubNetW4CleanAuxExDnNoTimeQdEma	2	0.0485	0.2033 ± 0.0180	0.1556 ± 0.0077
SubNetW4CleanAuxExDnOrhAbsQdEma	2	0.0397	0.1689 ± 0.0091	0.0967 ± 0.0041
SubNetW4CleanAuxExDnOrhQdEma	1	0.0409	0.1657	0.1052
SubNetW4CleanAuxExDnPfQdEma	2	0.0395	0.1599 ± 0.0160	0.1027 ± 0.0014
SubNetW4CleanAuxExDnQdEma	5	0.0390	0.1654 ± 0.0047	0.0957 ± 0.0038
SubNetW4CleanAuxExDnR12QdEma	1	0.0397	0.1760	0.1079
SubNetW4CleanAuxExDnR15QdEma	1	0.0407	0.1591	0.0997
SubNetW4CleanAuxExDnRcAcQdEma	5	0.0385	0.0761 ± 0.0047	0.0714 ± 0.0044
SubNetW4CleanAuxExDnRcK0AcQdEma	1	0.0384	0.0858	0.0779
SubNetW4CleanAuxExDnRcK1AcQdEma	1	0.0392	0.0714	0.0793
SubNetW4CleanAuxExDnRcK2AcQdEma	1	0.0382	0.0790	0.0730
SubNetW4CleanAuxExDnRcK3AcQdEma	1	0.0387	0.0756	0.0738
SubNetW4CleanAuxExDnRcK4AcQdEma	1	0.0388	0.0851	0.0749
SubNetW4CleanAuxExDnRcK5AcQdEma	1	0.0408	0.0885	0.0886
SubNetW4CleanAuxExDnRcK6AcQdEma	1	0.0408	0.0901	0.0824
SubNetW4CleanAuxExDnRcK7AcQdEma	1	0.0407	0.0930	0.0746
SubNetW4CleanAuxExDnRcK8AcQdEma	1	0.0416	0.0851	0.0850
SubNetW4CleanAuxExDnRcK9AcQdEma	1	0.0406	0.0881	0.0770
SubNetW4CleanAuxExDnRhoQdEma	2	0.0395	0.1583 ± 0.0018	0.0997 ± 0.0000
SubNetW4CleanAuxExDnSxQdEma	2	0.0398	0.1734 ± 0.0056	0.0989 ± 0.0050
SubNetW4CleanAuxExDnTpAuxQdEma	2	0.0397	0.1720 ± 0.0020	0.1054 ± 0.0042
SubNetW4CleanAuxExDnTpOrhFmQdEma	2	0.0404	0.1847 ± 0.0074	0.1016 ± 0.0079
SubNetW4CleanAuxExDnTpQdEma	1	0.0403	0.1826	0.1047
SubNetW4CleanAuxExDnTr10QdEma	2	0.0513	0.2309 ± 0.0587	0.1617 ± 0.0461
SubNetW4CleanAuxExDnTr20QdEma	2	0.0565	0.2636 ± 0.0494	0.1705 ± 0.0161
SubNetW4CleanAuxExDnW04QdEma	2	0.0399	0.1641 ± 0.0033	0.1029 ± 0.0070
SubNetW4CleanAuxExDnW05QdEma	1	0.0407	0.1772	0.0974
SubNetW4CleanAuxExDnW07QdEma	2	0.0410	0.1631 ± 0.0186	0.1087 ± 0.0071
SubNetW4CleanAuxExDnW1QdEma	1	0.0431	0.1868	0.1225
SubNetW4CleanAuxExGxQdEma	2	0.0393	0.1577 ± 0.0025	0.1049 ± 0.0031
SubNetW4CleanAuxExQdEma	2	0.0394	0.1761 ± 0.0052	0.1033 ± 0.0015

configuration	seeds	aggregate	15 mm low-E	30 mm low-E
SubNetW4CleanAuxGeoQdEma	1	0.0420	0.1804	0.1117
SubNetW4CleanAuxGeoQpQdEma	1	0.0414	0.1684	0.1083
SubNetW4CleanAuxNoTimeQdEma	2	0.0508	0.2559 ± 0.0001	0.1837 ± 0.0007
SubNetW4CleanAuxPhysOccExQdEma	2	0.0393	0.1755 ± 0.0019	0.1023 ± 0.0004
SubNetW4CleanAuxPhysOccQdEma	2	0.0408	0.1698 ± 0.0117	0.1051 ± 0.0077
SubNetW4CleanAuxPhysOccQpQdEma	5	0.0408	0.1681 ± 0.0079	0.0935 ± 0.0055
SubNetW4CleanAuxQdEma	5	0.0408	0.1726 ± 0.0041	0.1059 ± 0.0041
SubNetW4CleanAuxQpQdEma	1	0.0426	0.1747	0.1098
SubNetW4CleanAuxQuant	5	0.0427	0.1641 ± 0.0090	0.1119 ± 0.0043
SubNetW4CleanAuxSgnExDnQdEma	4	0.0395	0.1578 ± 0.0027	0.1028 ± 0.0037
SubNetW4CleanAuxTgateQdEma	2	0.0413	0.1744 ± 0.0001	0.1113 ± 0.0099
SubNetW5CleanAuxExDnQdEma	1	0.0393	0.1531	0.0932
SubNetW6CleanAuxExDnOrhQdEma	1	0.0404	0.1489	0.0953
SubNetW6CleanAuxExDnQdEma	3	0.0389	0.1189 ± 0.0027	0.0905 ± 0.0003
SubNetW6CleanAuxExDnR15QdEma	1	0.0399	0.1452	0.0945
SubNetW6CleanAuxExDnTpQdEma	1	0.0403	0.1328	0.0988
SubNetW6CleanAuxPhysOccQdEma	2	0.0407	0.1682 ± 0.0030	0.0964 ± 0.0001
SubNetW7CleanAuxExDnOrhQdEma	1	0.0402	0.1200	0.0867
SubNetW7CleanAuxExDnQdEma	5	0.0383	0.1131 ± 0.0065	0.0866 ± 0.0034
SubNetW7CleanAuxExDnR15QdEma	1	0.0394	0.1240	0.0893
SubNetW7CleanAuxExDnTpQdEma	1	0.0411	0.1427	0.0956
SubNetW8CleanAuxExDnB192RcQdEma	1	0.0429	0.0860	0.0837
SubNetW8CleanAuxExDnB48RcQdEma	1	0.0425	0.0899	0.0858
SubNetW8CleanAuxExDnCjOp2RcQdEma	1	0.0426	0.0837	0.0838
SubNetW8CleanAuxExDnCjOp5RcQdEma	1	0.0422	0.0865	0.0820
SubNetW8CleanAuxExDnCosRcQdEma	1	0.0424	0.0842	0.0817
SubNetW8CleanAuxExDnD256L6RcQdEma	1	0.1179	0.2983	0.2662
SubNetW8CleanAuxExDnD64L2RcQdEma	1	0.0463	0.1110	0.0996
SubNetW8CleanAuxExDnLrOp0001RcQdEma	1	0.0462	0.1076	0.0984
SubNetW8CleanAuxExDnLrOp0005RcQdEma	1	0.0434	0.0915	0.0829
SubNetW8CleanAuxExDnMmQdEma	3	0.0386	0.1049 ± 0.0023	0.0807 ± 0.0027
SubNetW8CleanAuxExDnQdEma	5	0.0386	0.1156 ± 0.0075	0.0779 ± 0.0024
SubNetW8CleanAuxExDnR15QdEma	5	0.0382	0.0997 ± 0.0109	0.0854 ± 0.0058
SubNetW8CleanAuxExDnR15RcRr01QdEma	3	0.0386	0.0725 ± 0.0037	0.0762 ± 0.0023
SubNetW8CleanAuxExDnRcFmQdEma	3	0.0390	0.0757 ± 0.0037	0.0745 ± 0.0020
SubNetW8CleanAuxExDnRcK0QdEma	3	0.0380	0.0781 ± 0.0025	0.0788 ± 0.0028
SubNetW8CleanAuxExDnRcK0Rr01QdEma	1	0.0388	0.0854	0.0788
SubNetW8CleanAuxExDnRcK1QdEma	3	0.0378	0.0780 ± 0.0048	0.0778 ± 0.0023
SubNetW8CleanAuxExDnRcK1Rr01QdEma	1	0.0398	0.0746	0.0797
SubNetW8CleanAuxExDnRcK2QdEma	3	0.0383	0.0720 ± 0.0020	0.0775 ± 0.0026
SubNetW8CleanAuxExDnRcK2Rr01QdEma	1	0.0386	0.0694	0.0798
SubNetW8CleanAuxExDnRcK3QdEma	3	0.0381	0.0787 ± 0.0018	0.0787 ± 0.0014
SubNetW8CleanAuxExDnRcK3Rr01QdEma	1	0.0385	0.0757	0.0761
SubNetW8CleanAuxExDnRcK4QdEma	3	0.0382	0.0813 ± 0.0065	0.0695 ± 0.0039
SubNetW8CleanAuxExDnRcK4Rr01QdEma	1	0.0381	0.0764	0.0761
SubNetW8CleanAuxExDnRcK5QdEma	3	0.0396	0.0836 ± 0.0063	0.0855 ± 0.0039
SubNetW8CleanAuxExDnRcK5Rr01QdEma	1	0.0406	0.0881	0.0875
SubNetW8CleanAuxExDnRcK6QdEma	3	0.0392	0.0805 ± 0.0073	0.0830 ± 0.0025
SubNetW8CleanAuxExDnRcK6Rr01QdEma	1	0.0402	0.0732	0.0795
SubNetW8CleanAuxExDnRcK7QdEma	3	0.0393	0.0924 ± 0.0041	0.0685 ± 0.0032
SubNetW8CleanAuxExDnRcK7Rr01QdEma	1	0.0402	0.0945	0.0783
SubNetW8CleanAuxExDnRcK8QdEma	3	0.0397	0.0713 ± 0.0054	0.0816 ± 0.0005
SubNetW8CleanAuxExDnRcK8Rr01QdEma	1	0.0406	0.0731	0.0789
SubNetW8CleanAuxExDnRcK9QdEma	3	0.0398	0.0760 ± 0.0021	0.0781 ± 0.0021
SubNetW8CleanAuxExDnRcK9Rr01QdEma	1	0.0410	0.0728	0.0810
SubNetW8CleanAuxExDnRcMmRr01QdEma	3	0.0388	0.0756 ± 0.0060	0.0754 ± 0.0018
SubNetW8CleanAuxExDnRcQdEma	5	0.0388	0.0711 ± 0.0059	0.0717 ± 0.0009
SubNetW8CleanAuxExDnRcRr01QdEma	5	0.0389	0.0700 ± 0.0033	0.0734 ± 0.0029
SubNetW8CleanAuxExDnRcTcQdEma	3	0.0413	0.0843 ± 0.0046	0.0786 ± 0.0016
SubNetW8CleanAuxExDnTpRcMmQdEma	1	0.0417	0.0975	0.0886
SubNetW8CleanAuxExDnTpRcQdEma	1	0.0409	0.0874	0.0817
SubNetW8CleanAuxGravExDnRcQdEma	2	0.0420	0.0842 ± 0.0045	0.0937 ± 0.0040

configuration	seeds	aggregate	15 mm low-E	30 mm low-E
SubNetW8CleanAuxMxrExDnRcQdEma	1	0.0654	0.2325	0.1485
SubNetW8CleanAuxPnetExDnRcQdEma	2	0.0410	0.0755 ± 0.0003	0.0832 ± 0.0052
SubNetW8CleanAuxSgnExDnRcQdEma	5	0.0392	0.0800 ± 0.0012	0.0850 ± 0.0036
SubNetW8CleanAuxStExDnQdEma	3	0.0442	0.1173 ± 0.0154	0.1015 ± 0.0039
SubNetW9CleanAuxExDnQdEma	1	0.0395	0.1074	0.0871

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